



Project Update:
H2 Float Water Level Monitoring System

KATHRYN DUNSTAN

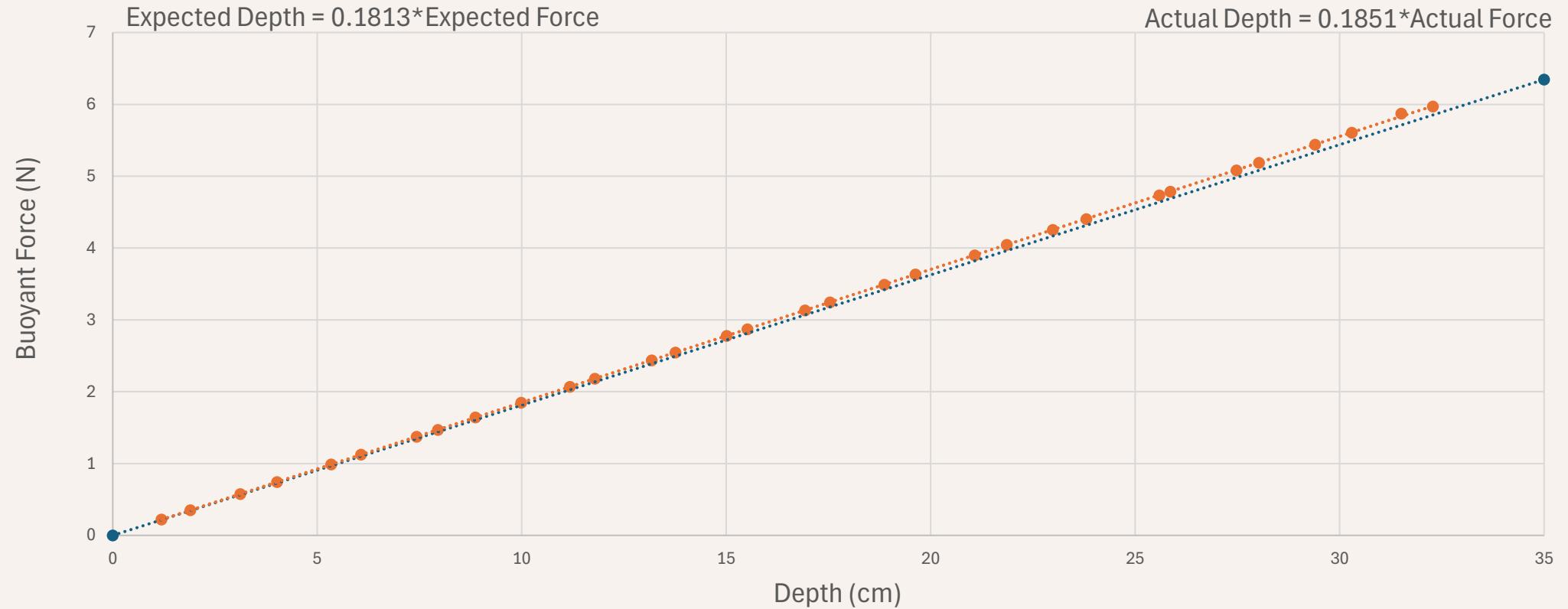
Progress Since Last Meeting

- Device build completed
- Calibration completed
- Initial testing completed
- Data collected
- Data analysis

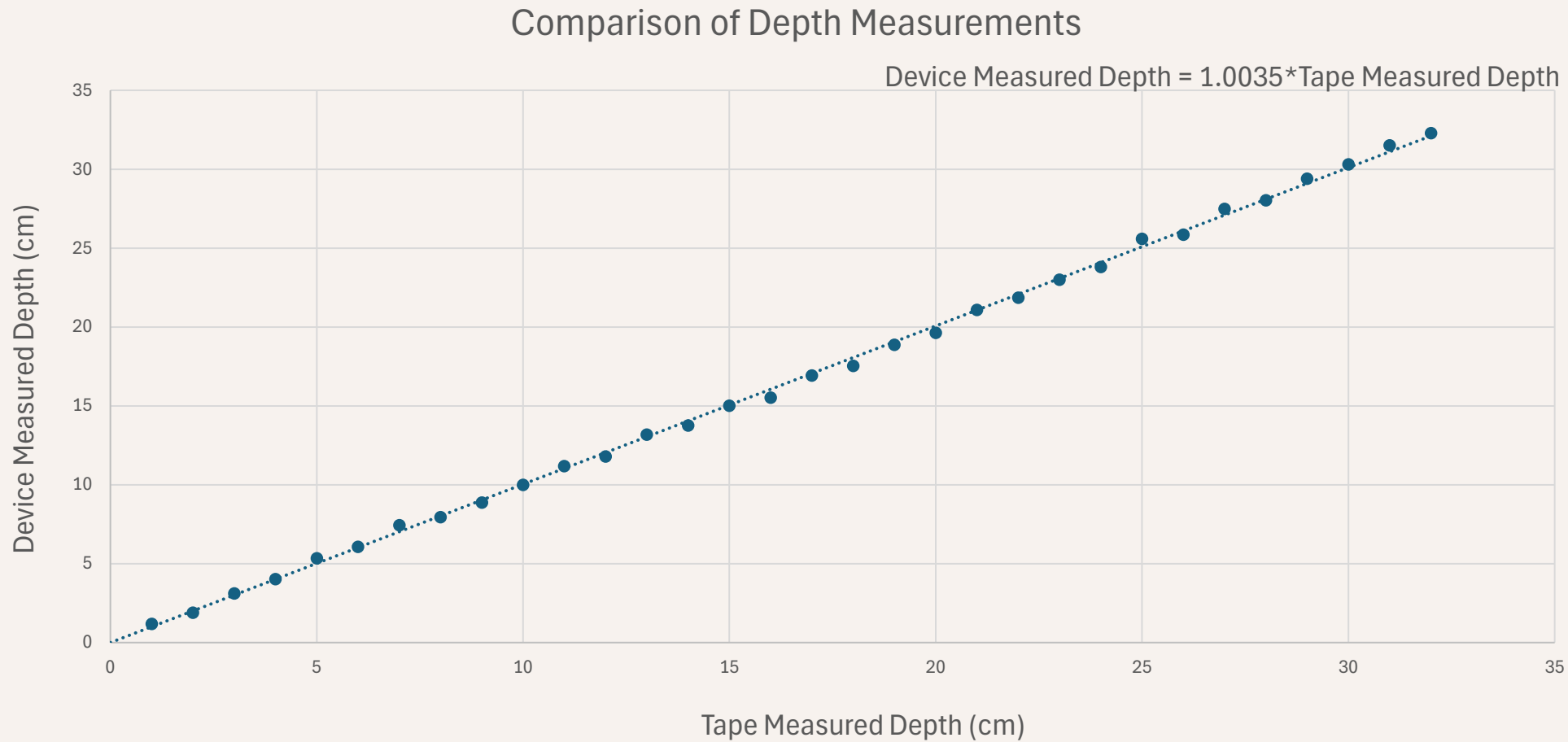


Buoyant Force Vs Water Depth

Expected Depth and Device Measured Depths
Based on Measured Buoyant Force



Device Measured Depth Vs Tape Measured Depth



Challenges

- Correcting systematic errors
 - Improve structure and support
 - Water containment has a flexing issue
 - Will pour leveling compound to combat this
 - ✓ Water movement skews data collection
 - ✓ Adjusted data collection timing
- Considerations
 - Temperature related expansion and contraction



Future Plans

- Setting up LoRa
- Moving into low power usage
 - LiPo Battery
 - Solar
- Saving to memory card
- Long term testing run
- Field test